

GAME THEORY

WHAT IS GAME THEORY?

- Mathematical tool to model the behavior and decision making of "people" interacting in a socio-economical context
- Every participant tries to optimize its private interest while its happiness is affected by the other participants' interests

=> PREDICT BEHAVIOUR

UNDERSTAND HOW COOPERATION, COMPETITION and CONFLICTS EMERGE

SOME EXAMPLES

PRISONER'S DILEMMA: Frank and Cora have been arrested and accused of a crime

They are questioned separately and they are proposed to confess

- if BOTH CONFESS, they both get 3 years
- if ONE CONFESS and the other DOESN'T the first is free, the second gets 5 years
- if BOTH DON'T CONFESS, they get 1 year

CORA'S CHOICE

		CONFESS		NOT CONF	
FRANK'S CHOICE	C	-3	0	-3	-5
	NC	-5	-1	0	-1

FRANK'S YEARS

CORA'S YEARS

		CORA			
		C	NC		
FRANK	C	-3, -3	0, -5	FRANK	CORA
	NC	-5, 0	-1, -1		

DOMINANT STRATEGY: CONFESS gives the lowest number of years, considering what the other prisoner would do

=> RATIONAL CHOICE

But they could have gotten less years if they didn't confess!

UNCOOPERATIVE GAME ←

- FRANK and COEN did not communicate
- We play the game only once
- We take into account only prison years

EXAMPLE FOR PRISONERS DILEMMA: competition between firms
escalation of re-armament or disarmament
agreement of countries on emissions

LET'S CHANGE THE INTERESTS

		COEN	
		C	NC
FRANK	C	-3, -3	-2, -5
	NC	-5, -2	-1, -1

Maybe we include some moral considerations or fear of future repercussions

NO more dominant strategy!

=> COORDINATION GAME

NASH EQUILIBRIUM: if one of the two could change its choice, it wouldn't, there is no incentive in changing

However one of the EE is preferable...

CHICKEN GAME: two cars drive towards each other
 every driver can decide to swerve to
 avoid crashing with the other
 The first one who swerve is the "CHICKEN"
 and the other one is brave
 If none of them swerve, they crash and die
 If both swerve, they don't put much consideration
 from their peers

		PLAYER 2	
		SWERVE	NO SWERVE
PLAYER 1	S	(0, 0)	(1, -1)
	NS	(-1, 1)	(-10, -10)

(Death)

ANTI-COORDINATION GAME:
 both the players have
 a preferred equilibrium

POSSIBLE STRATEGY: convince
 the other player, you
 are not going to swerve!

REAL LIFE EXAMPLE: Americans bomb Russia,
 Russians have a secret device
 that, in case of an attack,
 will cause the final
 counteroffensive
 ⇒ Russian's steering wheel is
 blocked but they haven't
 told the Americans
 - Dr. STRANGELOVE, Stanley KUBRICK

ZERO-SUM GAMES: two players flip two coins,
 If they get 2 tails or 2 heads, P1 wins
 otherwise P2 wins

		H	T
P1	H	1, -1	-1, 1
	T	-1, 1	1, -1

NO EQUILIBRIA!

Typical of zero-sum
 games

⇒ MIXED STRATEGY:
 EQ is given by 50%
 heads, 50% tails

BEST RESPONSE

BEST RESPONSE: action that max our player's gain in response to the other player's action

NASH EQUILIBRIUM: a pair of strategies such that one is the best response for the other and vice versa

THEOREM: every game with a finite number of players and a finite number of strategies admits a Nash equilibrium

=> MIXED EQUILIBRIUM!

ROCK-PAPER-SCISSORS

	R	P	S
R	0,0	-1,1	1,-1
P	1,-1	0,0	-1,1
S	-1,1	1,-1	0,0

There are too much info, the game is zero-sum!

0	-1	1
1	0	-1
-1	1	0

p prob of playing ROCK

q prob of playing PAPER

$1-p-q$ prob of playing SCISSORS

now I

=> EXPECTED PAYOFF PLAYING ROCK: $0p - 1q + 1(1-p-q)$ (1)

PAPER: $1p + 0q - 1(1-p-q)$ (2)

SCISSORS: $-1p + q + 0(1-p-q)$ (3)

=> if we are at the eq these should all be the same!

$$(1) = (2) \quad \begin{cases} -q + 1 - p - q = p - 1 + p + q \end{cases}$$

$$(2) = (3) \quad \begin{cases} p - 1 + p + q = -p + q \end{cases}$$

$$\begin{cases} -3q - 3p + 2 = 0 \\ 3p = 1 \end{cases}$$

$$\begin{cases} p = \frac{1}{3} \\ q = \frac{1}{3} \end{cases}$$

NON-ZERO-SUM BIMATRIX GAMES

$$\begin{bmatrix} (1,3) & (0,2) \\ (2,-1) & (0,1) \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 0 \\ 2 & 0 \end{bmatrix}$$

$$B = \begin{bmatrix} 3 & 2 \\ -1 & 1 \end{bmatrix}$$

PLAYER A plays TOP with prob p_T

BOTTOM

$$p_B = 1 - p_T$$

$$\Rightarrow p = \begin{bmatrix} p_T \\ p_B \end{bmatrix}$$

p_1
 p_2

PLAYER B plays LEFT with prob q_L

RIGHT

$$q_R$$

$$\Rightarrow q = \begin{bmatrix} q_L \\ q_R \end{bmatrix}$$

q_1
 q_2

EXPECTED OUTCOME PLAYER A:

$$\begin{aligned} & 1 \cdot p_T \cdot q_L + 0 \cdot p_T \cdot q_R + 2 \cdot p_B \cdot q_L + 0 \cdot p_B \cdot q_R = \\ & = \begin{bmatrix} p_T & p_B \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 2 & 0 \end{bmatrix} \begin{bmatrix} q_L \\ q_R \end{bmatrix} = p^T A q \end{aligned}$$

EXPECTED OUTCOME PLAYER B:

$$\begin{aligned} & 3 \cdot p_T \cdot q_L + 2 \cdot p_T \cdot q_R + (-1) \cdot p_B \cdot q_L + 1 \cdot p_B \cdot q_R = \\ & = \begin{bmatrix} p_T & p_B \end{bmatrix} \begin{bmatrix} 3 & 2 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} q_L \\ q_R \end{bmatrix} = p^T B q \end{aligned}$$

(MIXED-STRATEGY) NASH EQUILIBRIUM: a pair (p^*, q^*) is a NE of a bimatrix game when both players minimize their cost if

$$\begin{aligned} & (p^*)^T A q^* \leq p^T A q^* \quad \forall p \\ & (p^*)^T A q^* \leq (p^*)^T A q \quad \forall q \end{aligned}$$

BATTLE of the SEXES or 'PIZZA vs SUSHI' EXERCISE / THIS IS WEIRD - CHECK OTHER

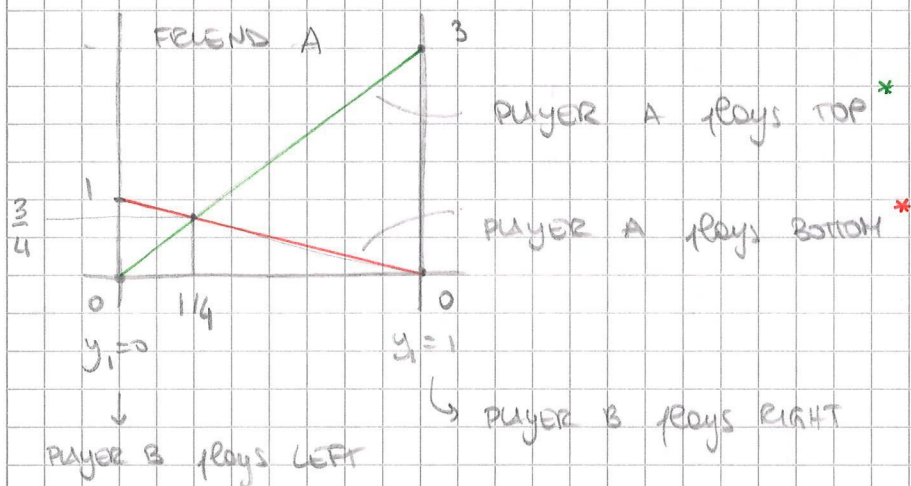
The BATTLE of the SEXES is a (very stereotypical) coordination game where a MAN and a WOMAN wants to meet in the evening but the man wants to go to a PIZZA FIGHT and the woman to the BAR.

Since I don't like this example, I use the PIZZA vs SUSHI example

Two friends want to go out for dinner, one ~~wants~~ wants to have PIZZA while the other wants SUSHI, however they would like to have dinner together

		FRIEND B			
		P	S		
FRIEND A	P	3, 1	0, 0	$A = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}$	$B = \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix}$
	S	0, 0	1, 3		

GRAPHICAL METHOD



(0, 1) (1, 0)

(0, 0) (1, 3)

$$\frac{x-0}{1} = \frac{y-1}{-1}$$

$$\frac{x-0}{1-0} = \frac{y-0}{3-0}$$

$$-x = y - 1$$

$$3x = y$$

$$y = 1 - x *$$

$$y = 3x *$$

$$\begin{cases} y = 1 - x \\ y = 3x \end{cases}$$

$$3x = 1 - x$$

$$x = 1/4$$

$$y = 3/4$$

MIXED STRATEGY,
PIZZA with prob 1/4

"hoffmann"

ALICE vs BOB (random names for Player A and Player B)

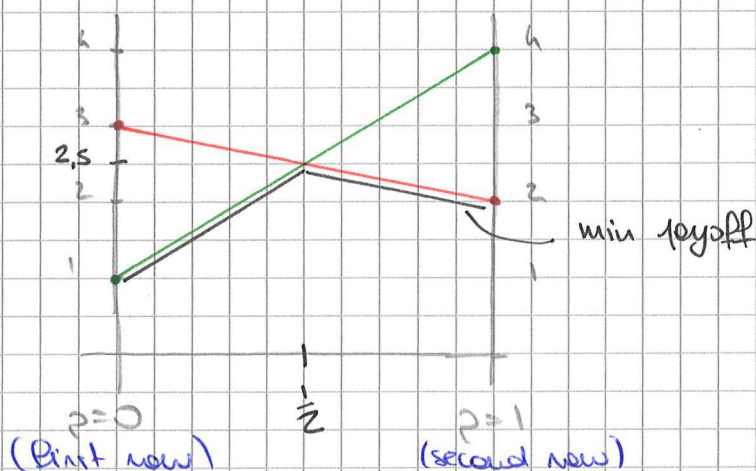
		B	
		b_1	b_2
A	a_1	1	3
	a_2	4	2
		$1-p$	p

ALICE'S PAYOFF MATRIX = BOB'S LOSS MATRIX
SHE WANTS TO MAXIMIZE HE WANTS TO MINIMIZE

ALICE'S EXPECTED PAYOFF:

$$b_1 \quad 1-p + 4p = 1+3p^*$$

$$b_2 \quad 3(1-p) + 2p = 3-p^*$$



$$\begin{cases} y = 1+3p \\ y = 3-p \end{cases}$$

$$3-p = 1+3p$$

$$4p = 2$$

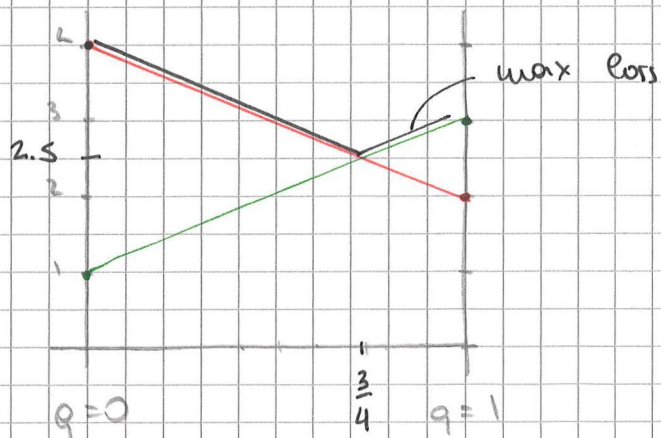
$$p = \frac{1}{2} \quad y = \frac{5}{2} = 2.5$$

BOB PICKS b_1 with prob $1-p$ and b_2 with prob p

BOB'S EXPECTED PAYOFF:

$$a_1 \quad 1-p + 3p = 1+2p$$

$$a_2 \quad 4(1-p) + 2p = 4-2p^*$$



$$\begin{cases} y = 1+2p \\ y = 4-2p \end{cases}$$

$$1+2p = 4-2p$$

$$4p = 3$$

$$p = \frac{3}{4} \quad y = \frac{10}{4} = 2.5$$

Is it a coincidence that Alice and Bob have the same payoff / EV value?

NO!

MIN-MAX THEOREM: every $n \times m$ matrix S has a VAW, i.e. there is a unique number v st

$$\max_x \min_y x S y^T = v = \min_y \max_x x S y^T$$

NON-ZERO SUM GAMES

		B	
		b_1	b_2
A	a_1	1, 6	3, -4
	a_2	4, 0	2, 5
		$1-p$	p

A	a_1	B	b_1
	a_2		b_2
B	b_1	A	a_1
	b_2		a_2

NO PURE EQ

MIXED STRATEGY: A

$$\begin{aligned} 1-p + 3p &= 1+2p \\ 4(1-p) + 2p &= 4-2p \end{aligned}$$

$$\begin{aligned} a_1 & 1+2p = 4-2p \\ a_2 & p = 3/4 \end{aligned}$$

B

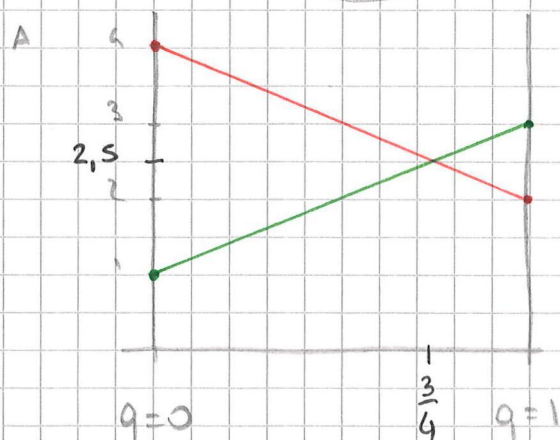
$$\begin{aligned} 6(1-p) \\ -4(1-p) + 5p &= -4+9p \end{aligned}$$

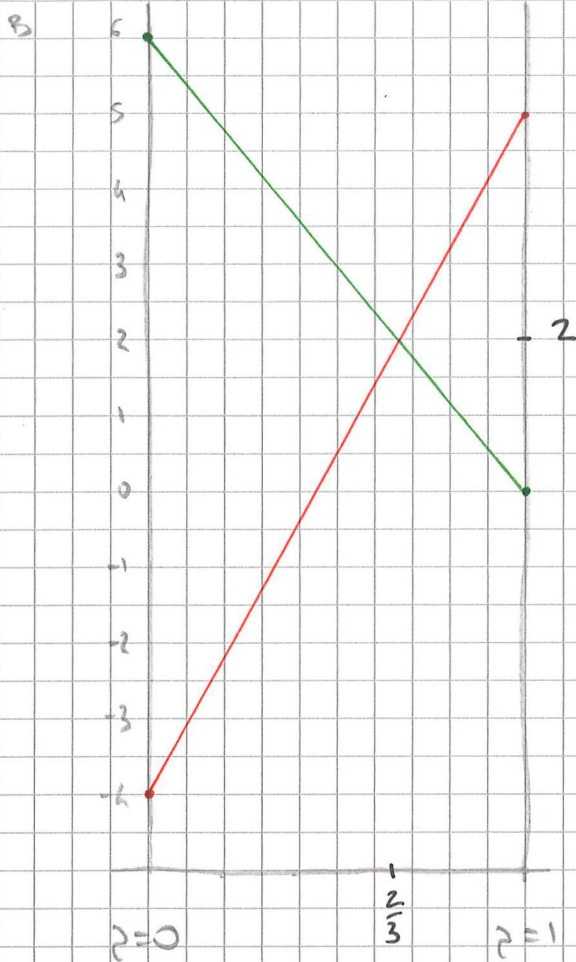
$$\begin{aligned} b_1 & -4+9p = 6-6p \\ b_2 & -10 = -15p \\ p &= \frac{2}{3} \end{aligned}$$

$$\begin{aligned} \text{VAWE of EQ: } y &= 1+2p = \frac{10}{4} = 2.5 \quad (A) \\ y &= 6(1-p) = 2 \quad (B) \end{aligned}$$

REMARK: in non-zero games the payoff depends on what the other player is doing!

We can use the graphical method here as well





REMARK: in this case we obtain the value of the expected payoff for B, but the mixed strategy for A

IDEA: GOAL IS TO MAKE OTHER PLAYER INDIFFERENT [BECAUSE NASH EQ] SO P1 CHOOSES p IN ORDER TO MAKE P2 INDIFFERENT $\Rightarrow$ VALUE FOR P1 STRATEGY FOR P2

